# Complex Sinusoids

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# Key Concepts

1) The real part of a complex sinusoid is a cosine wave and the imaginary part is a sine wave.

2) A complex sinusoid $x(t) = Ae^{j\Omega t+\phi}$ can be visualized in the complex plane as a vector of length $A$ that is rotating at a rate of $\Omega$ radians per second and has angle $\phi$ relative to the real axis at time $t = 0$.

    a) The projection of the complex sinusoid onto the real axis is the cosine $A\cos(\Omega t + \phi)$.

    b) The projection of the complex sinusoid onto the imaginary axis is the cosine $A\sin(\Omega t + \phi)$.

3) The output of a linear time invariant system in response to a complex sinusoid input is a complex sinusoid of the of the same frequency. The system only changes the amplitude and phase of the input complex sinusoid.

    a) Arbitrary input signals are represented as weighted sums of complex sinusoids of different frequencies.

    b) Then, the output is a weighted sum of complex sinusoids where the weights have been multiplied by the complex gain of the system.

4) Complex sinusoids also simplify algebraic manipulations.

5) Real sinusoids can be expressed as a sum of a positive and negative frequency complex sinusoid.

6) The concept of negative frequency is not physically meaningful for real data and is an artifact of using complex sinusoids.

    a) A complex sinusoid of positive frequency is represented in the complex plane as a vector rotating in a counter clockwise direction.

    b) A complex sinusoid of negative frequency is represented in the complex plane as a vector rotating in a clockwise direction.

c) The sign of the frequency captures the relationship between the real and imaginary components of the complex sinusoid.

# Video Screenshots

# Complex sinusoids have both real and imaginary parts!

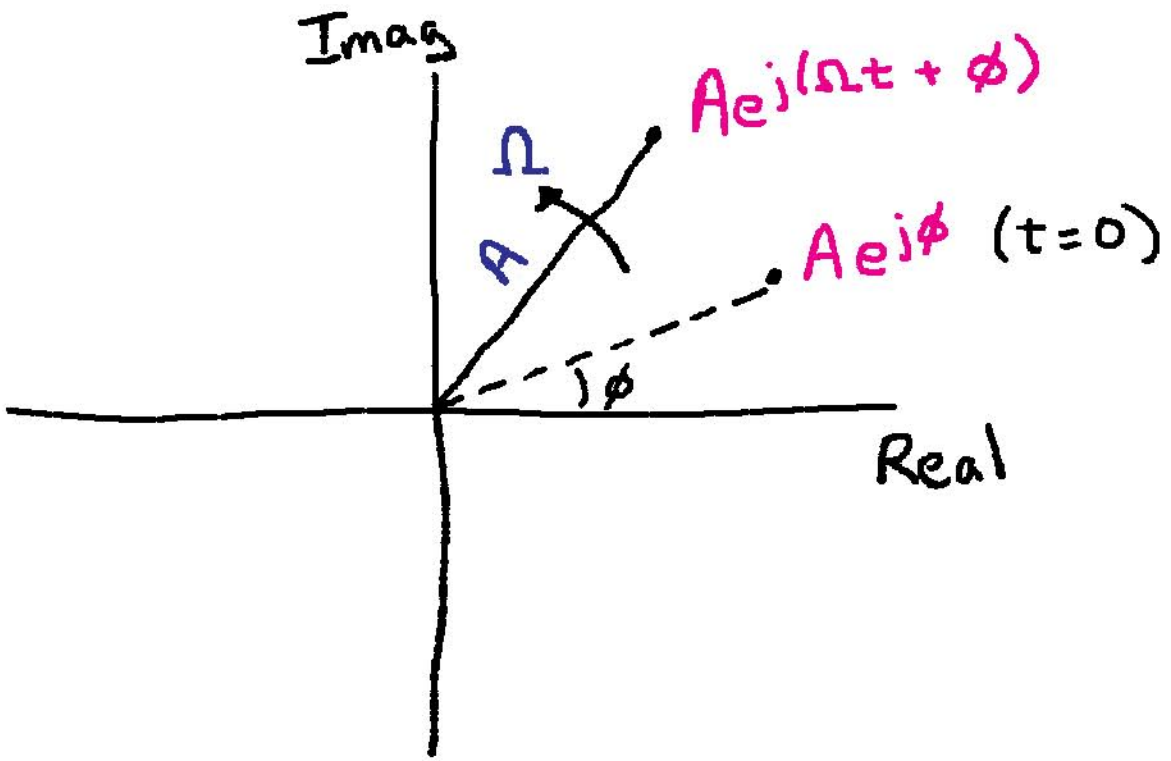
"polar coords"

$$x(t) = A e^{j(\Omega t + \phi)}$$

$$= A\cos(\Omega t + \phi) + jA\sin(\Omega t + \phi)$$

real imaginary

by Euler's formula

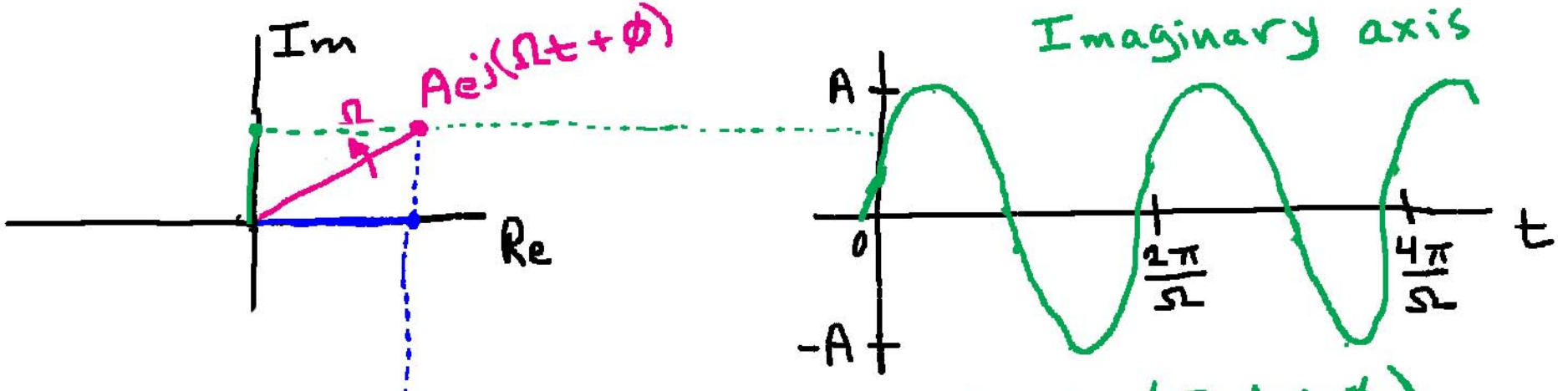


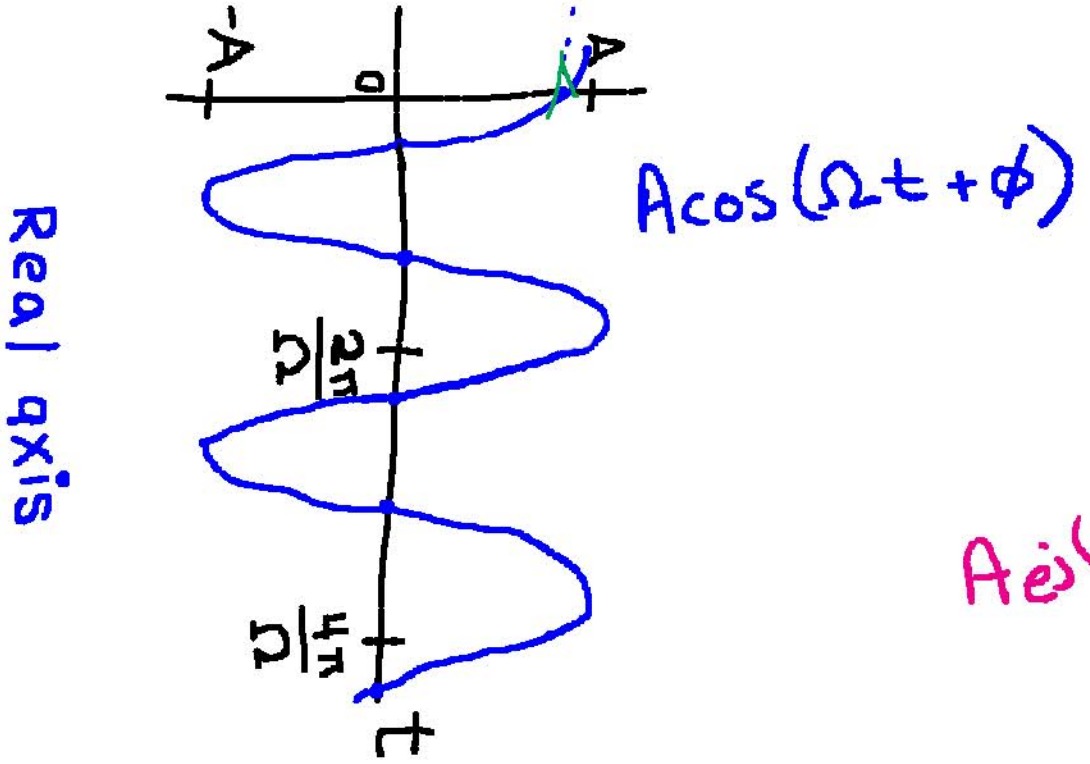

Length of vector: A

Rate of rotation: $\Omega$ $\frac{\text{rads}}{\text{sec}}$

(frequency)

Starting phase: $\phi$ rads



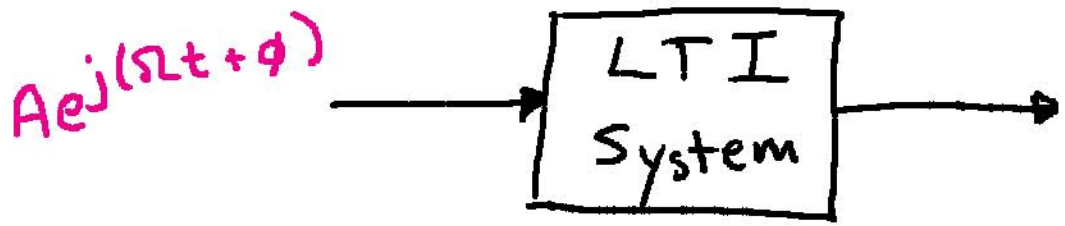




$$Ae^{j(\Omega t+\phi)} = A\cos(\Omega t+\phi) + jA\sin(\Omega t+\phi)$$

# Why complex sinusoids?



1) Linear, time invariant systems behave very simply




$C(\Omega)$ is a complex number that depends only on $\Omega$

Use as "building blocks"

2) Simplicity of complex algebra

If $B = Ae^{j\phi}$, $B \cdot e^{j\Omega t} = Ae^{j(\Omega t + \phi)}$ multiplication

$B\sin(\Omega t + \phi)$ phase inside transcendental function

$= B\sin\phi\cos(\Omega t) + B\cos\phi\sin(\Omega t)$

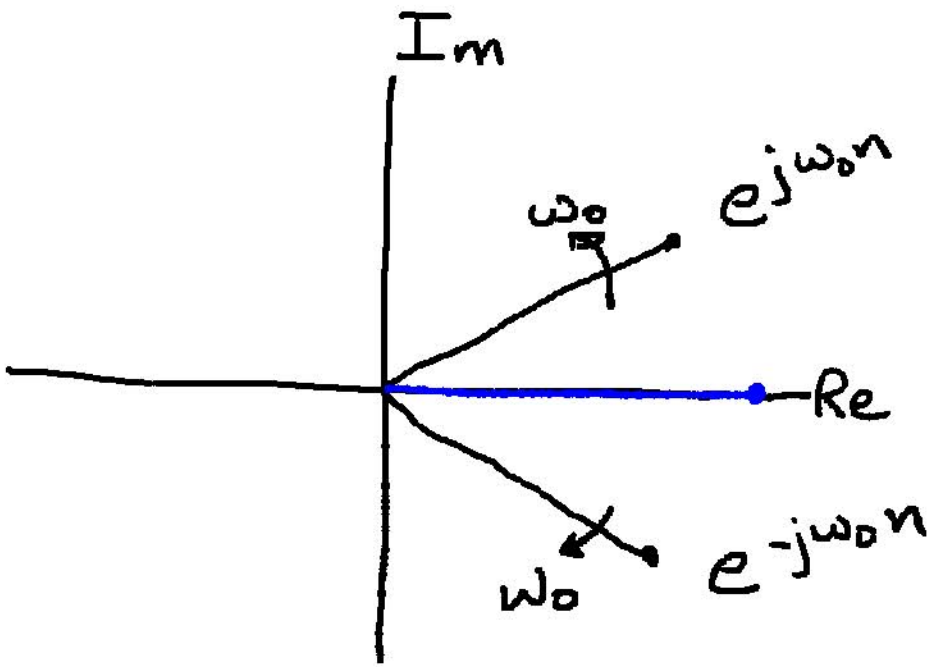
multiplication, but two real sinusoids

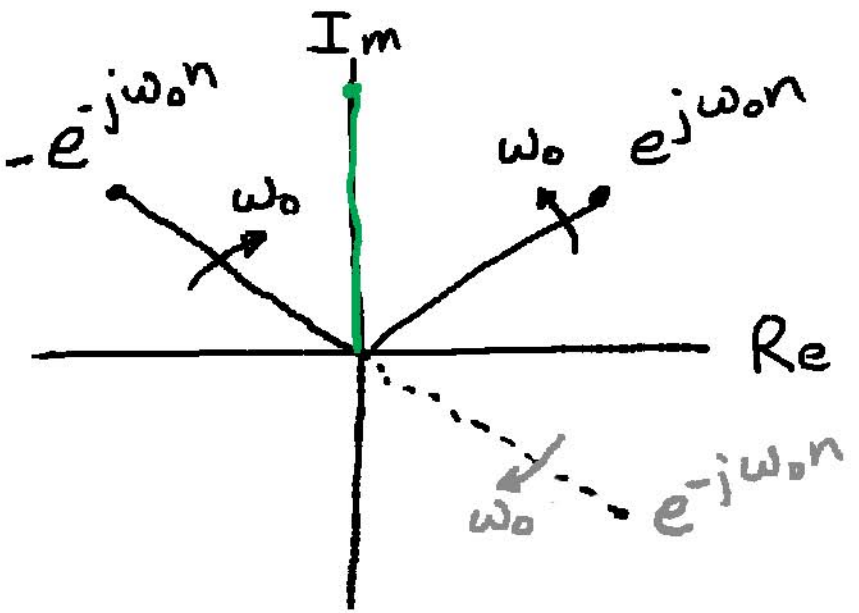
# Useful identities



$$\cos(\omega_0 n) = \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}$$

$$\sin(\omega_0 n) = \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}$$

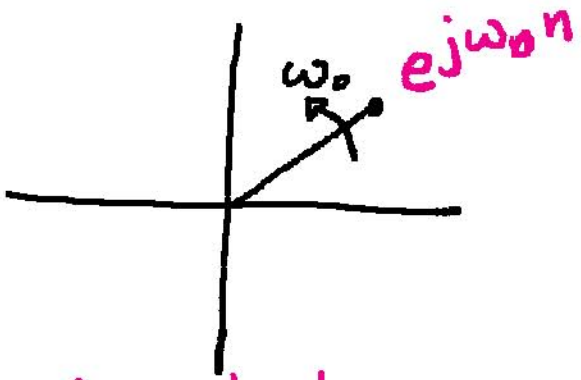


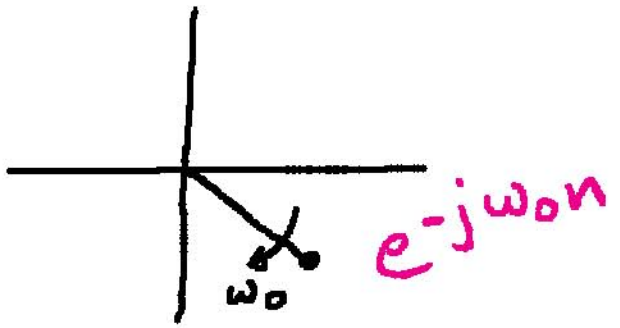

Real sinusoid: combination of + and − frequency complex sinusoids

# Negative frequency ?

- Artifact of using complex sinusoids
- Not meaningful for real-valued sinusoids




counter clockwise




clockwise

$e^{j\omega_0 n} = \cos(\omega_0 n) + j\sin(\omega_0 n)$

$e^{-j\omega_0 n} = \cos(\omega_0 n) - j\sin(\omega_0 n)$

- Sign captures relationship between real and imaginary components

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